Digital Image Processing 6:
Segmentation- Point, Line, Edge Detection

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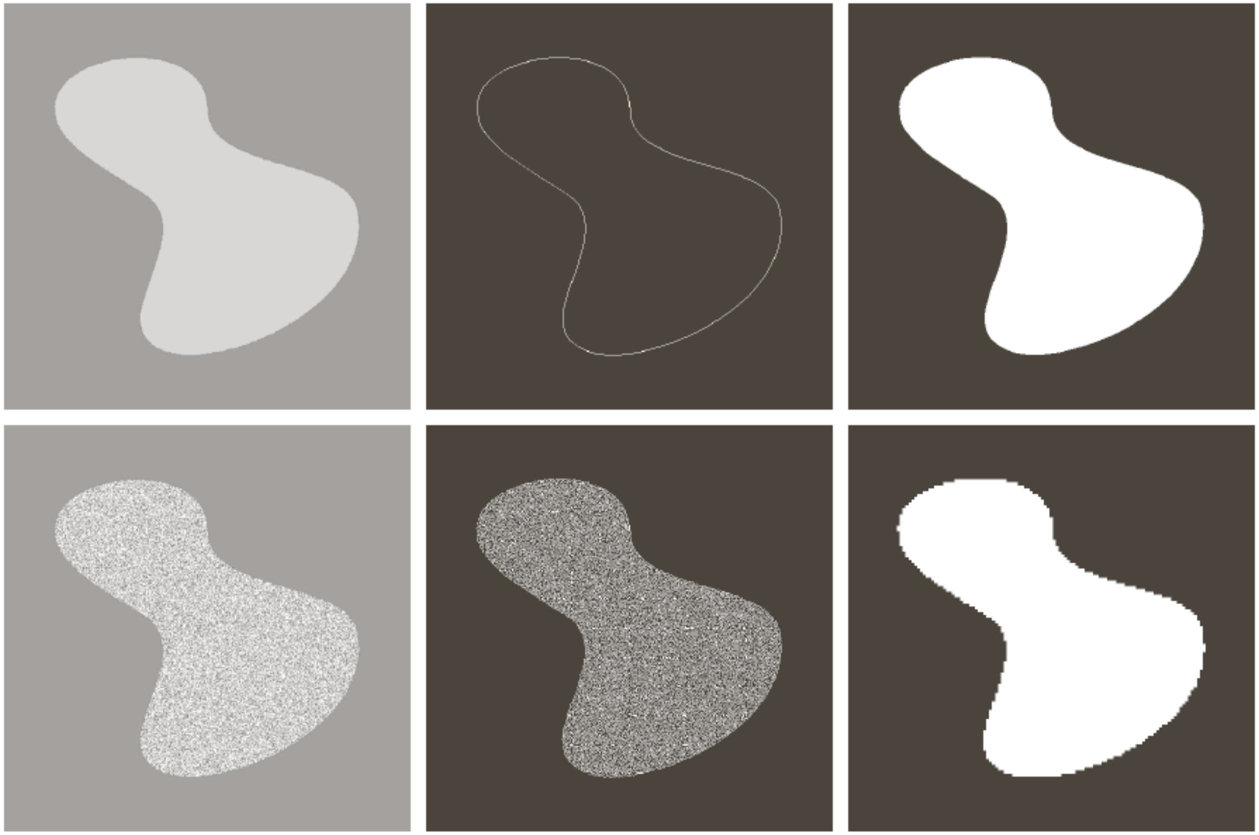
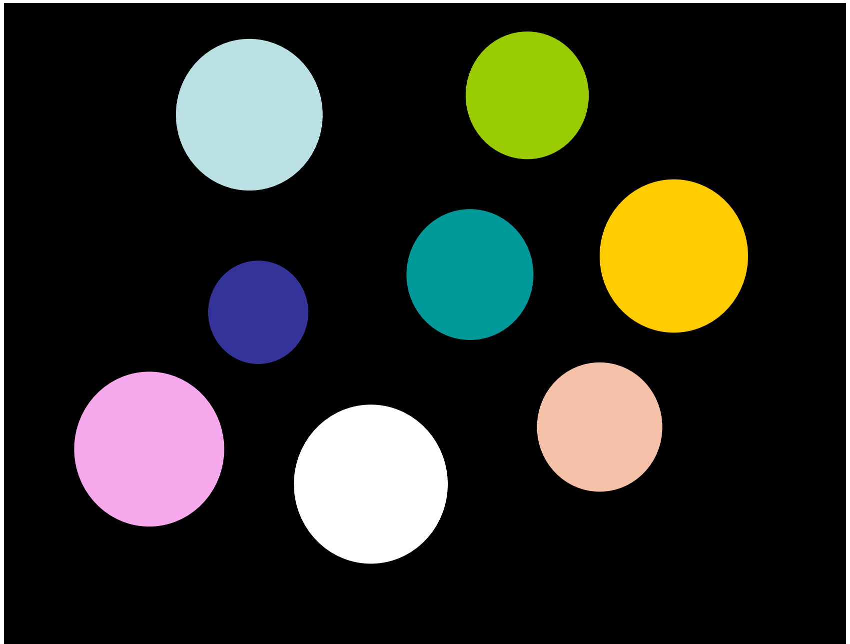
Contents

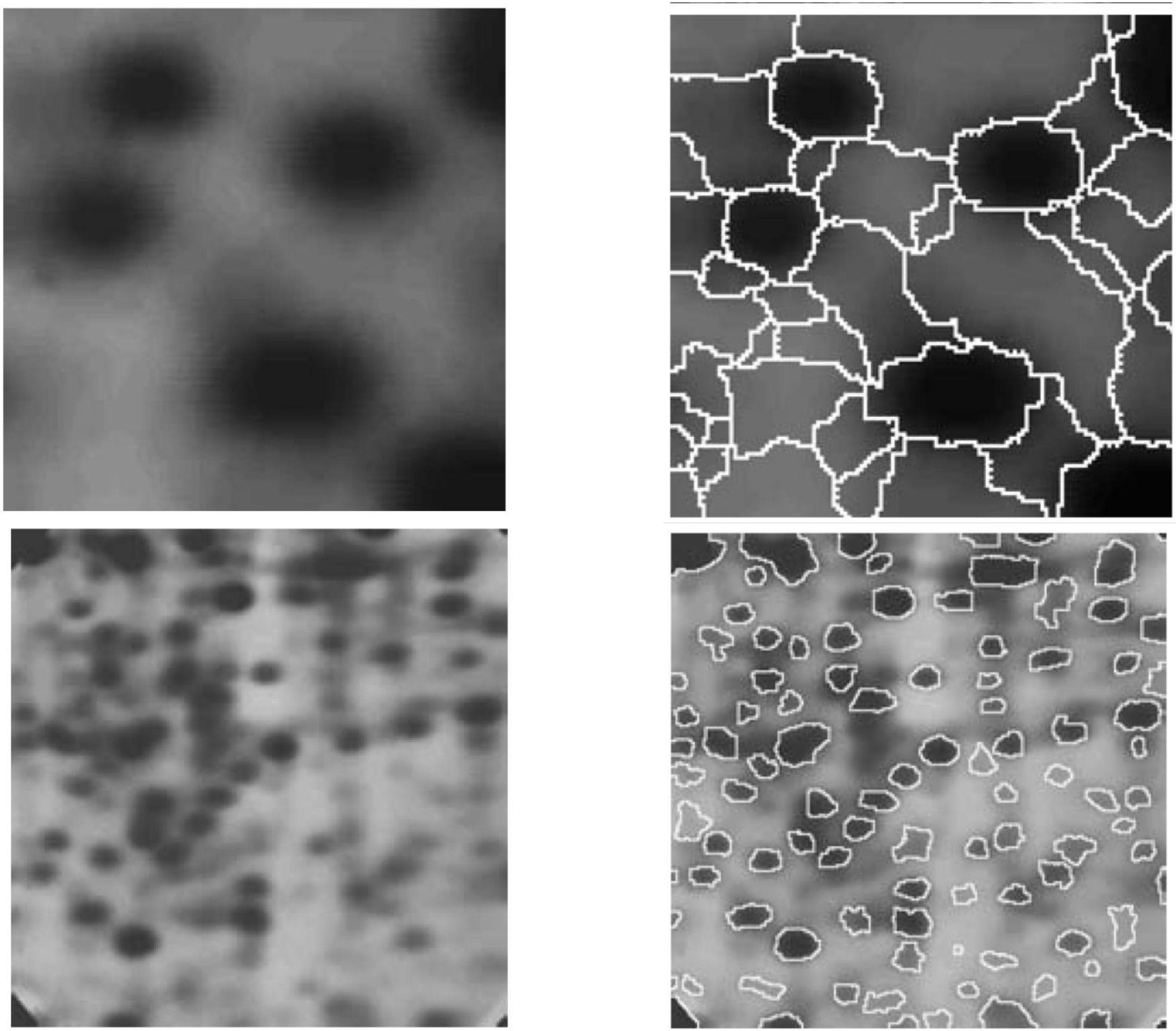
1. What is segmentation
2. Finding points, lines and edges



Segmentation

- *Definition:* attempts to partition the pixels of an image into groups that strongly correlate with the objects in an image.
- Typically the first step in any automated computer vision application.





- Segmentation Methods

Based on two properties of gray-level image values

1. Discontinuity
    - Edge detection
2. Similarity
    - Thresholding
    - Region growing / splitting / merging

Detection Discontinuities

- Three basic types of grey level discontinuities are tended to look for in digital images:
  - Points
  - Lines
  - Edges
- Method: using masks and correlation
- Knowledge from the previous chapters

- Detector (points, lines, and edges)!!

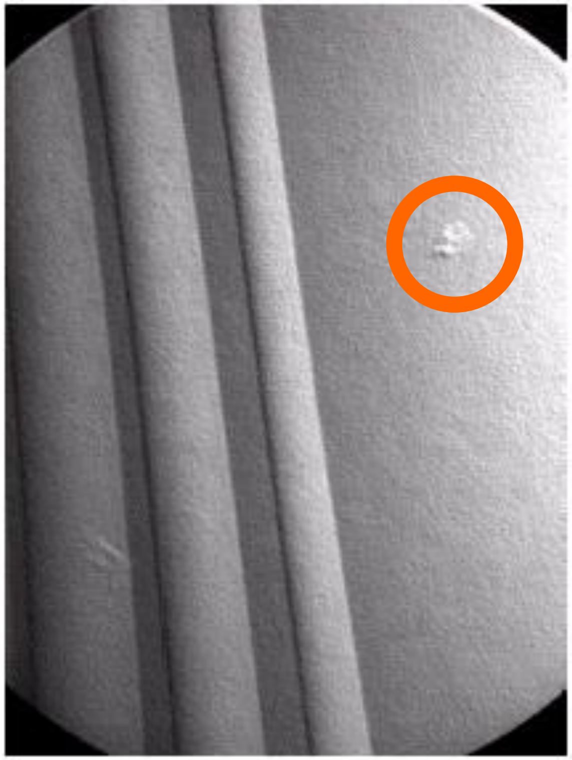
Detection Discontinuities--Points

Point detection can be achieved simply using the mask below:

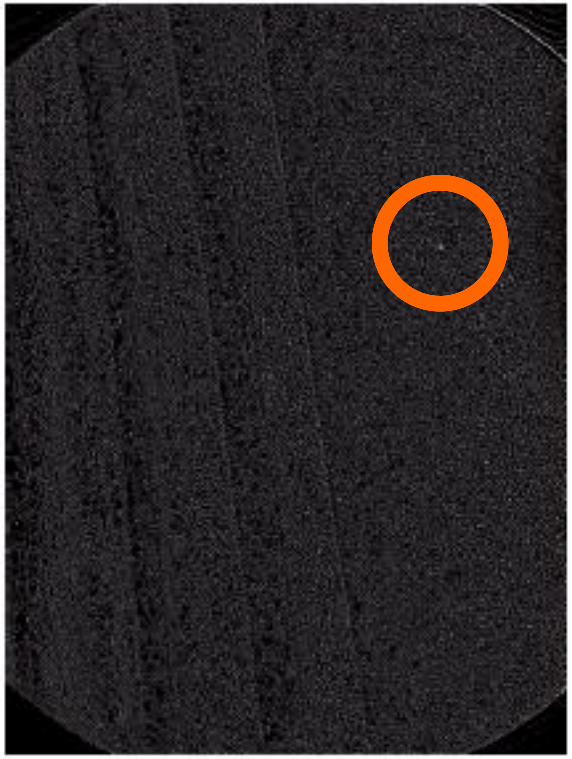
–1	–1	–1
–1	8	–1
–1	–1	–1

Points are detected at those pixels in the subsequent filtered image that are above a set threshold

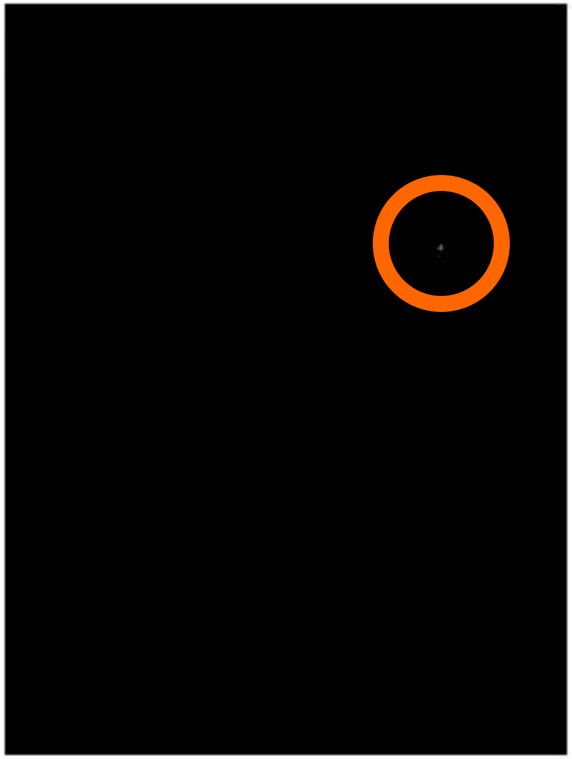
Detection Discontinuities--Points



X-ray image of a turbine blade



Result of point detection



Result of thresholding

Detection Discontinuities--Lines

The masks below will extract lines that are one pixel thick and running in a particular direction.

−1	−1	−1
2	2	2
−1	−1	−1

Horizontal

−1	−1	2
−1	2	−1
2	−1	−1

$+45°$

−1	2	−1
−1	2	−1
−1	2	−1

Vertical

2	−1	−1
−1	2	−1
−1	−1	2

$-45°$

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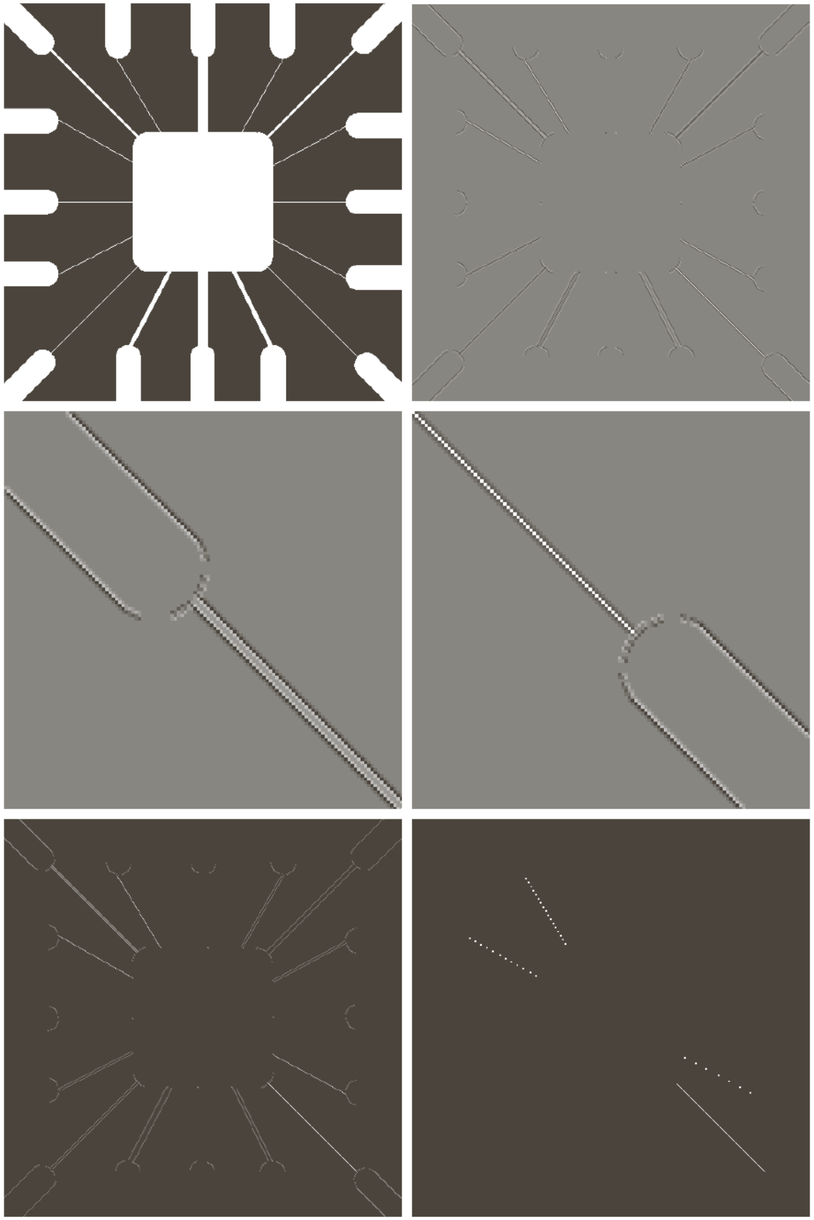
Detection Discontinuities--Lines

Binary image of a wire bond mask

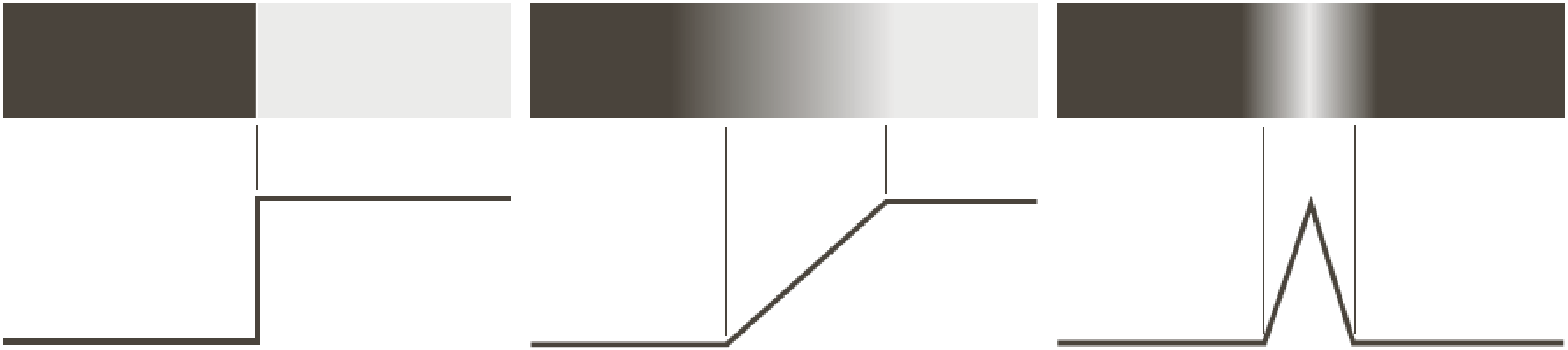


After processing with -45° line detector

Result of thresholding filtering result

Detection Discontinuities--Edges

Definition: An edge is a set of connected pixels that lie on the boundary between two regions



a b c

FIGURE 10.8 From left to right, models (ideal representations) of a step, a ramp, and a roof edge, and their corresponding intensity profiles.

Detection Discontinuities--Edges

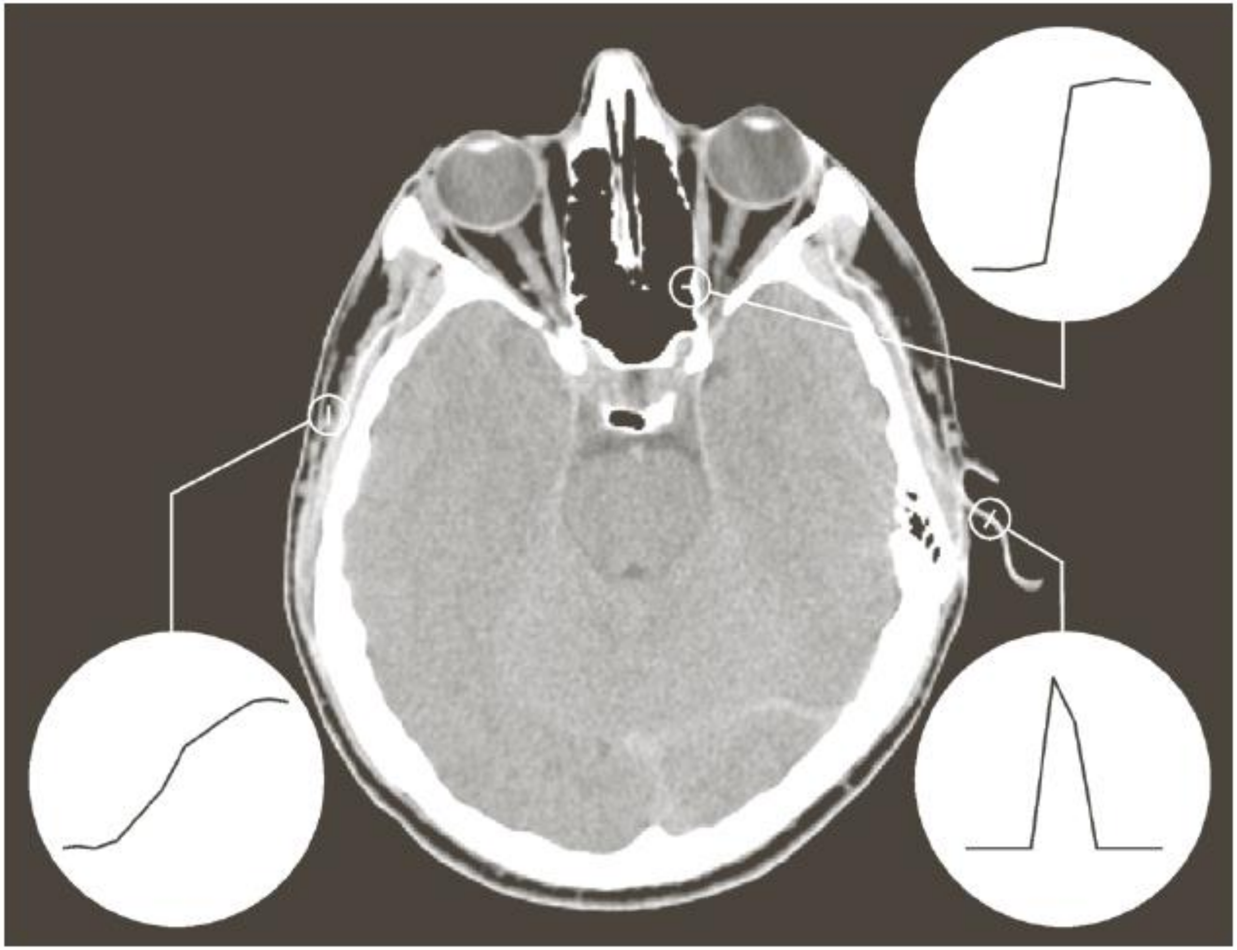


FIGURE 10.9 A 1508 × 1970 image showing (zoomed) actual ramp (bottom, left), step (top, right), and roof edge profiles. The profiles are from dark to light, in the areas indicated by the short line segments shown in the small circles. The ramp and "step" profiles span 9 pixels and 2 pixels, respectively. The base of the roof edge is 3 pixels. (Original image courtesy of Dr. David R. Pickens, Vanderbilt University.)

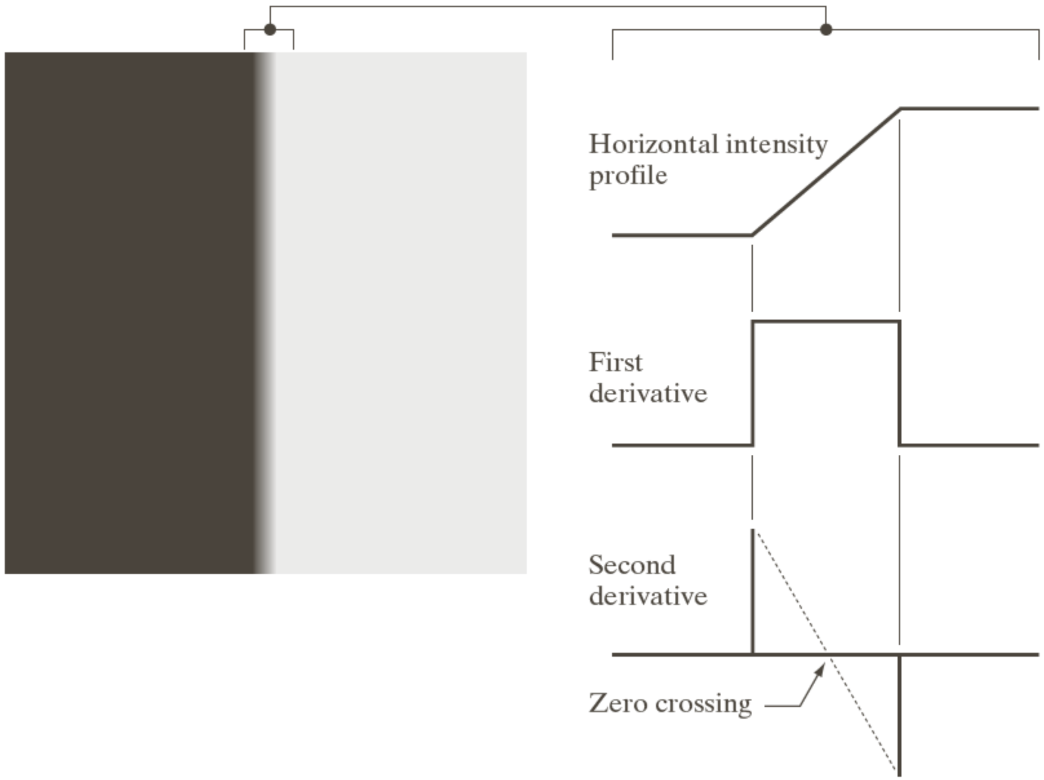
Detection Discontinuities--Edges

Averaging smoothens an image. Given that averaging is analogous to integration, local changes in intensity can be detected using derivatives.

Derivatives and edges:

✓1st derivative tells us where an edge is.

✓2nd derivative can be used to show edge direction



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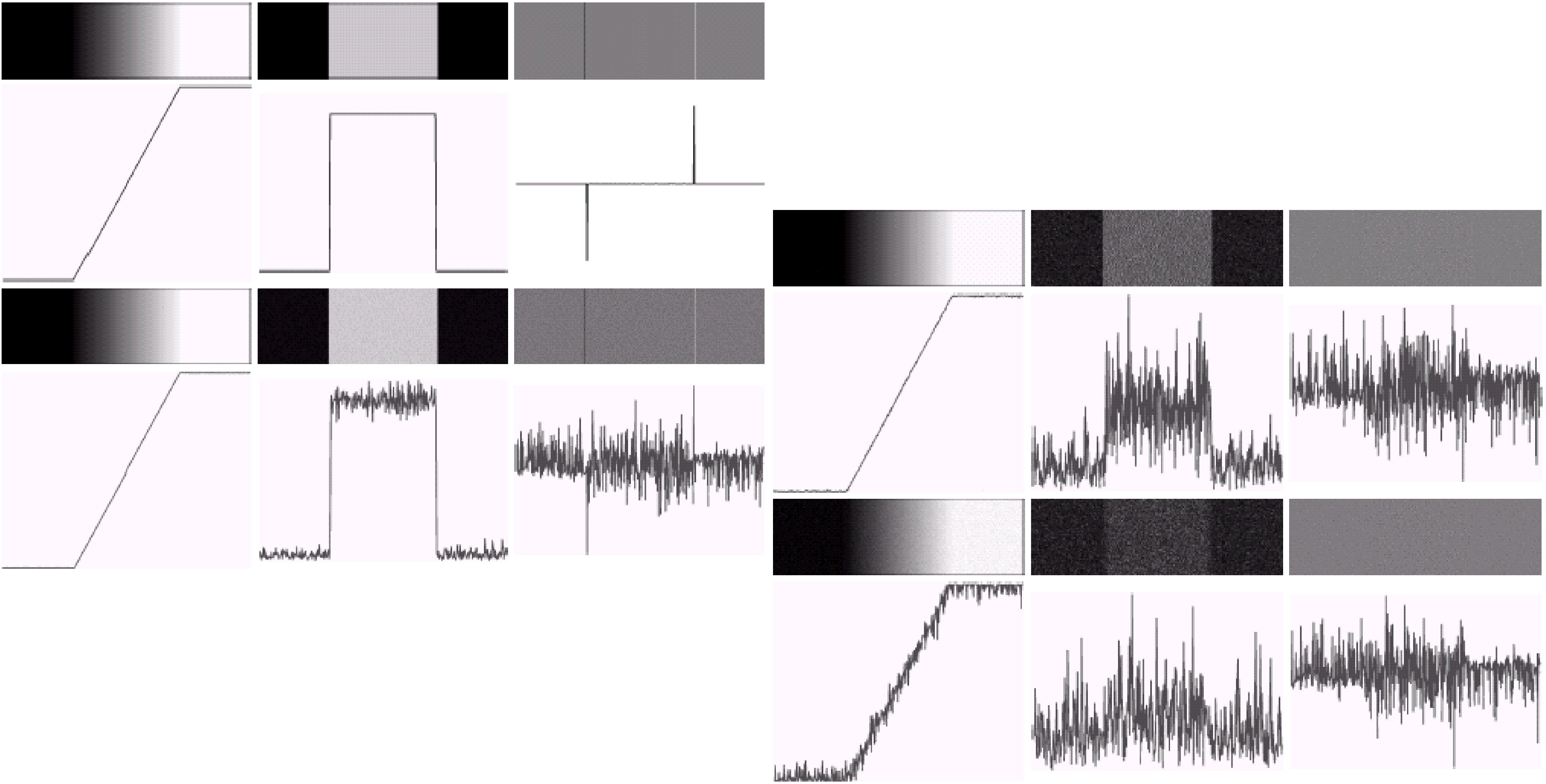
Detection Discontinuities--Edges

Derivative based edge detectors are **extremely sensitive** to noise



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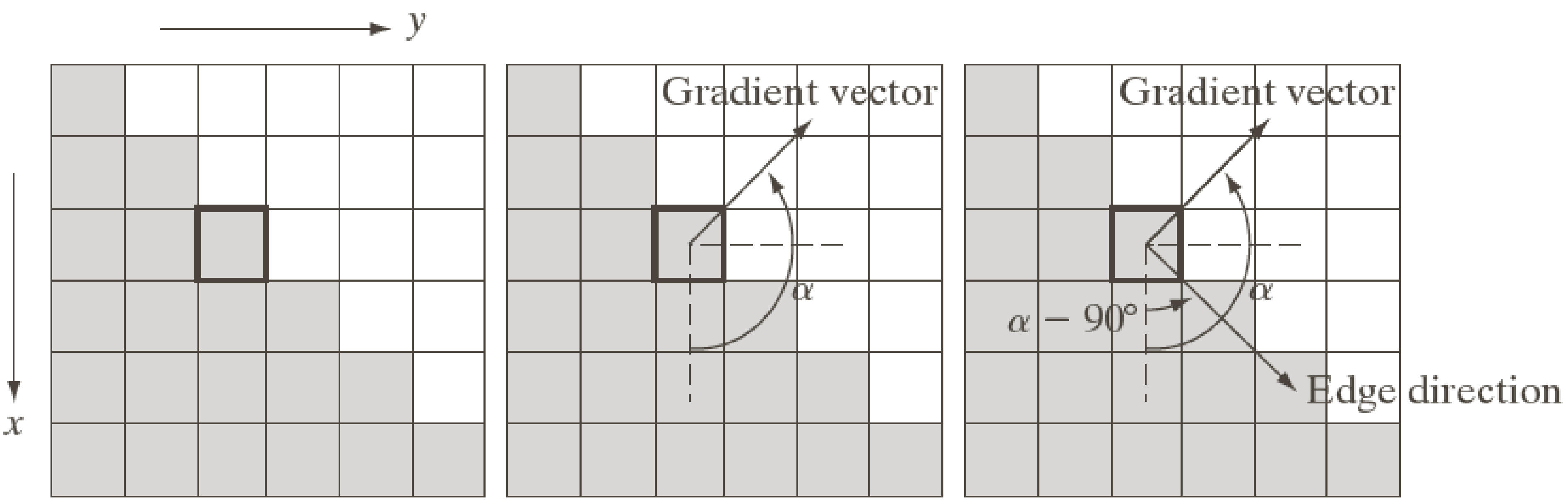
Detection Discontinuities--Edges

Gradient



Basic Edge Detectors

A 3*3 region of an image and masks for edge detection

z_1	z_2	z_3
z_4	z_5	z_6
z_7	z_8	z_9

−1	0
0	1

0	−1
1	0

Roberts

Center!

Basic Edge Detectors

A 3*3 region of an image and masks for edge detection

z_1	z_2	z_3
z_4	z_5	z_6
z_7	z_8	z_9

−1	−1	−1
0	0	0
1	1	1

−1	0	1
−1	0	1
−1	0	1

Prewitt

Basic Edge Detectors

A 3*3 region of an image and masks for edge detection

z_1	z_2	z_3
z_4	z_5	z_6
z_7	z_8	z_9

−1	−2	−1
0	0	0
1	2	1

−1	0	1
−2	0	2
−1	0	1

Sobel

Original Image

Horizontal Gradient Component

Vertical Gradient Component

Combined Edge Image

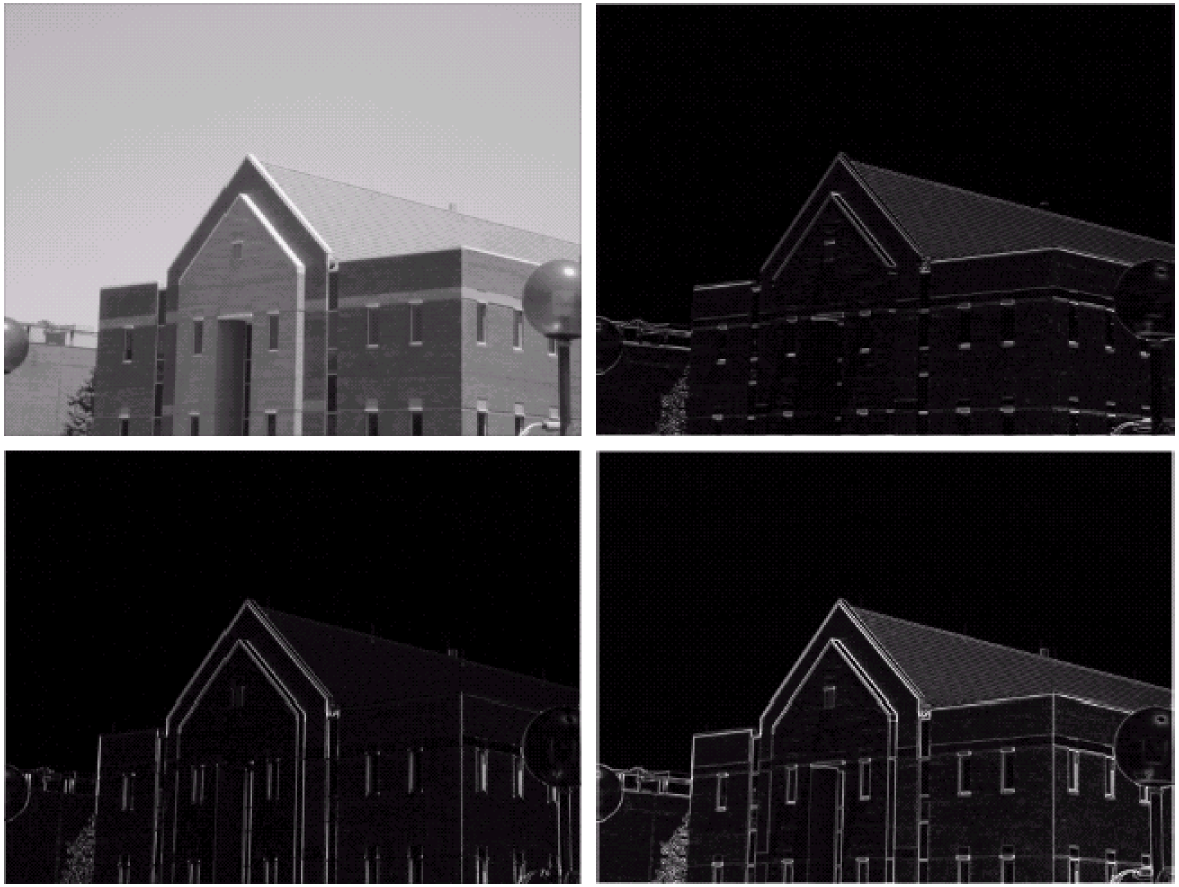
Edge Detection Problems

•Edge detection results in **too much detail**, for example, the brickwork in the previous example.

•One way to overcome this is to smooth images first.

Edge Detection Problems

- Smoothing using 5*5 average filter, and then calculate vertical and horizontal gradients.
- But all the edges are weakened.



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Edge Detection Problems

- Diagonal edge detection

Laplacian Edge Detection(2^{nd} order)

- The 2^{nd}-order derivative based Laplacian filter.

0	−1	0
−1	4	−1
0	−1	0

−1	−1	−1
−1	8	−1
−1	−1	−1

- Laplacian filter is too sensitive to noise.
- Usually it is combined with a smoothing Gaussian filter.

Laplacian of Gaussian[LoG] !

Laplacian Edge Detection

Then, we have Marr-Hildreth edge detector

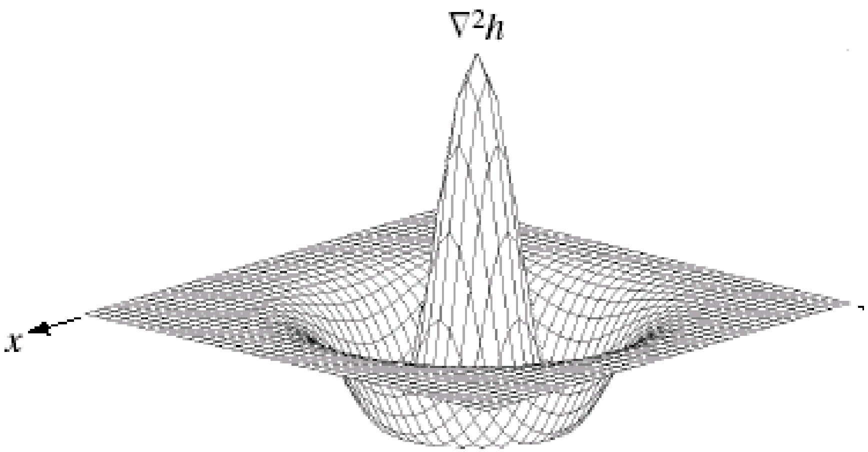
Laplacian of Gaussian (Mexican hat) filter uses:

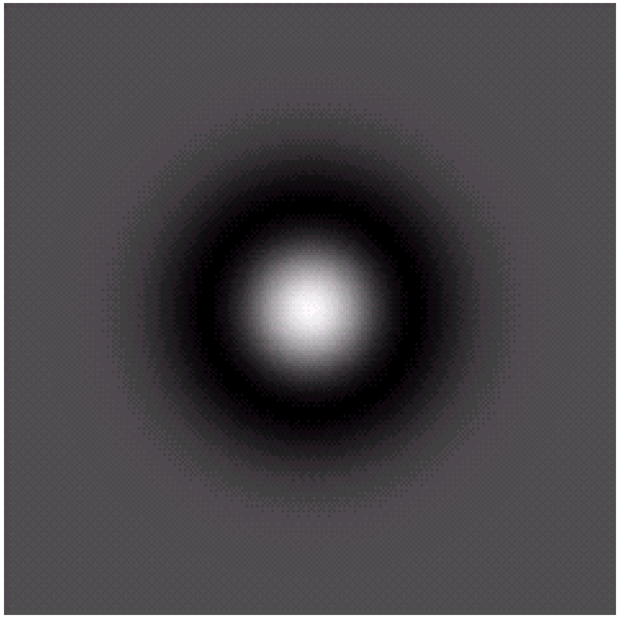
- Step one: Gaussian for noise removal;
- Step two: Laplacian for edge detection.

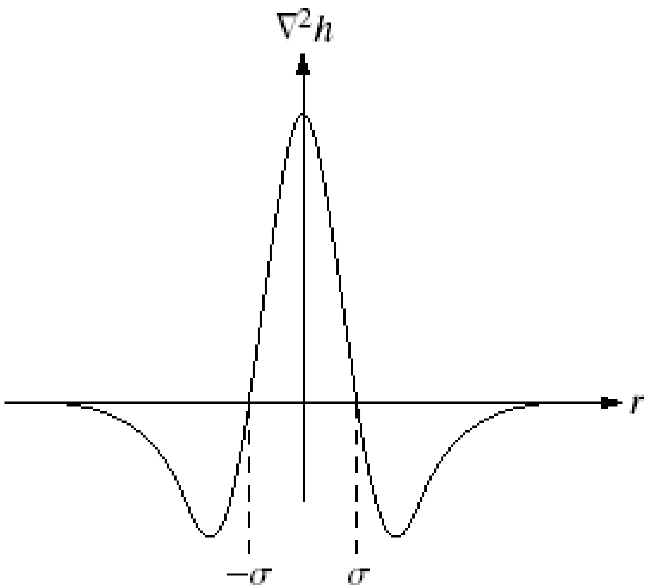
Algorithm:

1. Filter the input image with n*n Gaussian lowpass filter for smoothing.
2. Compute the Laplacian of the image from step 1 using m*m mask.
3. Find the zero crossings of the image from step 2.

Laplacian Edge Detection







0	0	−1	0	0
0	−1	−2	−1	0
−1	−2	16	−2	−1
0	−1	−2	−1	0
0	0	−1	0	0

a b
c d

FIGURE 10.21 (a) Three-dimensional plot of the *negative* of the LoG. (b) Negative of the LoG displayed as an image. (c) Cross section of (a) showing zero crossings. (d) 5×5 mask approximation to the shape in (a). The negative of this mask would be used in practice.

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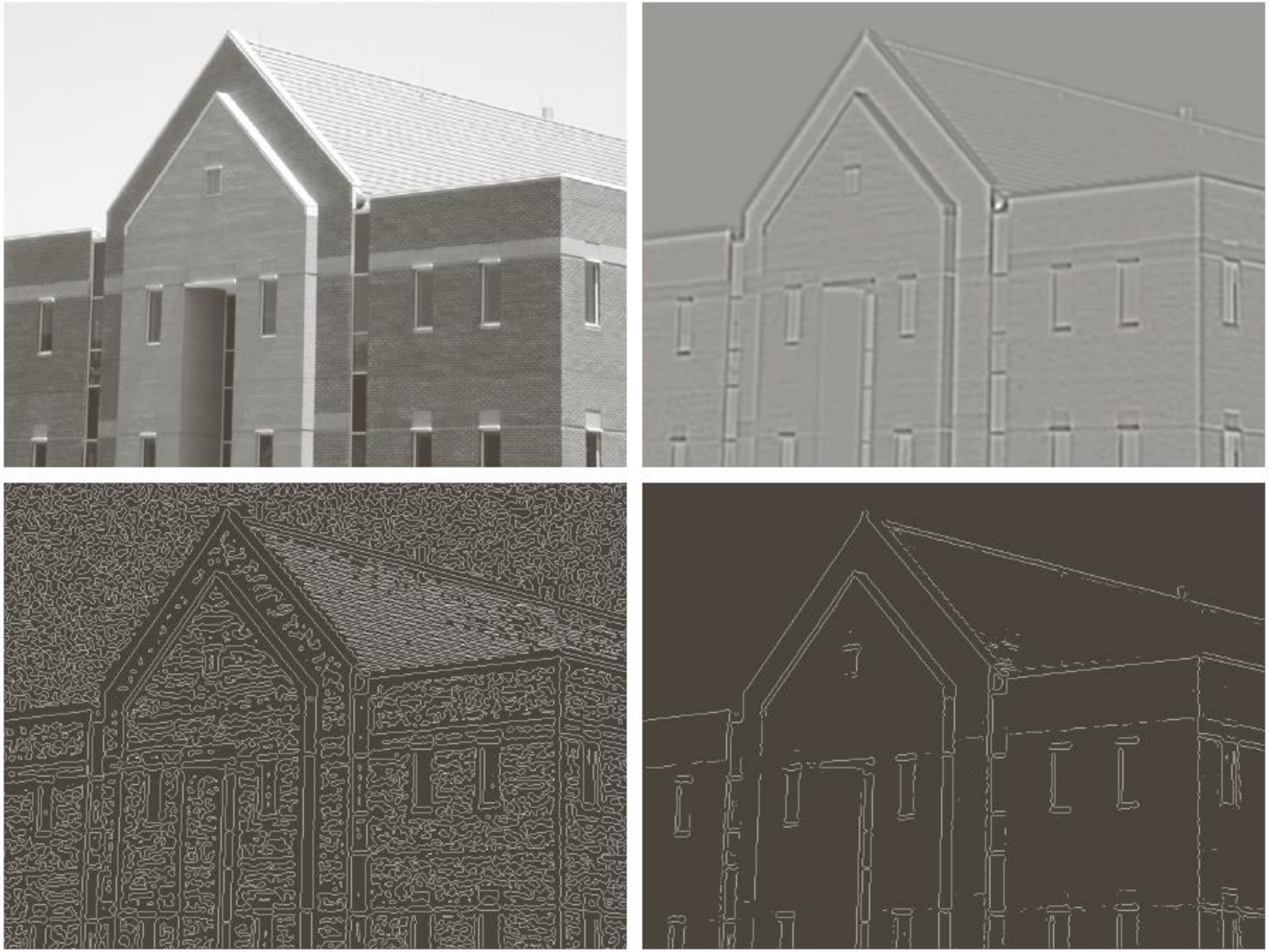
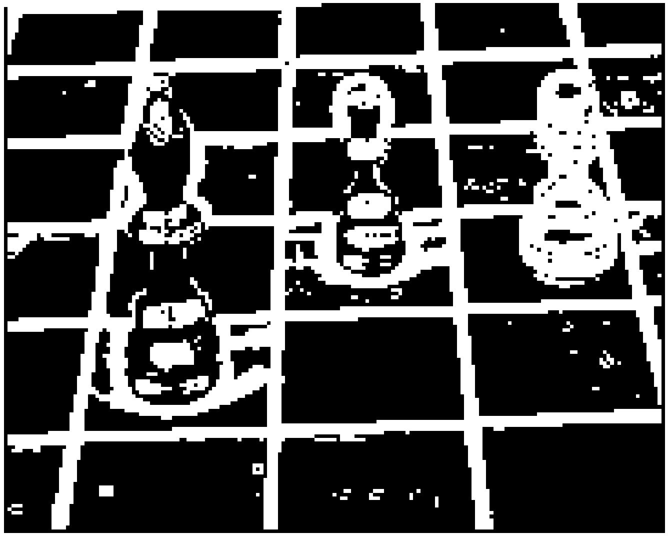
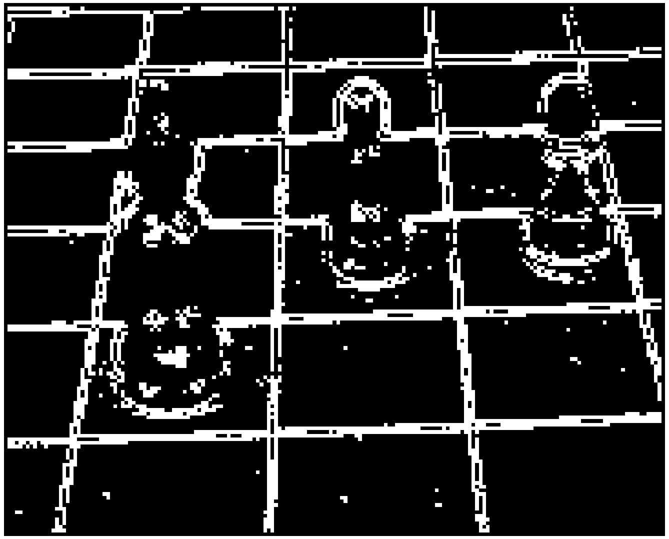


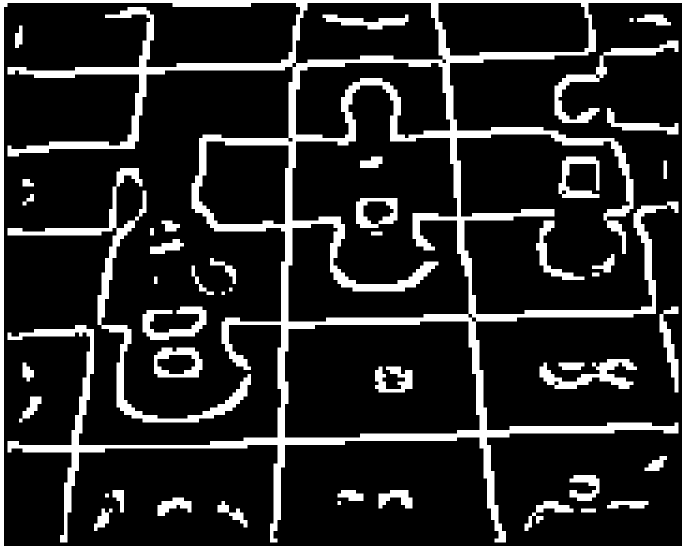
a	b
c	d

FIGURE 10.22
(a) Original image of size 834×1114 pixels, with intensity values scaled to the range [0, 1]. (b) Results of Steps 1 and 2 of the Marr-Hildreth algorithm using $\sigma = 4$ and $n = 25$. (c) Zero crossings of (b) using a threshold of 0 (note the closed-loop edges). (d) Zero crossings found using a threshold equal to 4% of the maximum value of the image in (b). Note the thin edges.



Sobel
Threshold=50





Laplacian
Threshold=30

Marr-Hildreth
Std=3.0

Canny Edge Detection

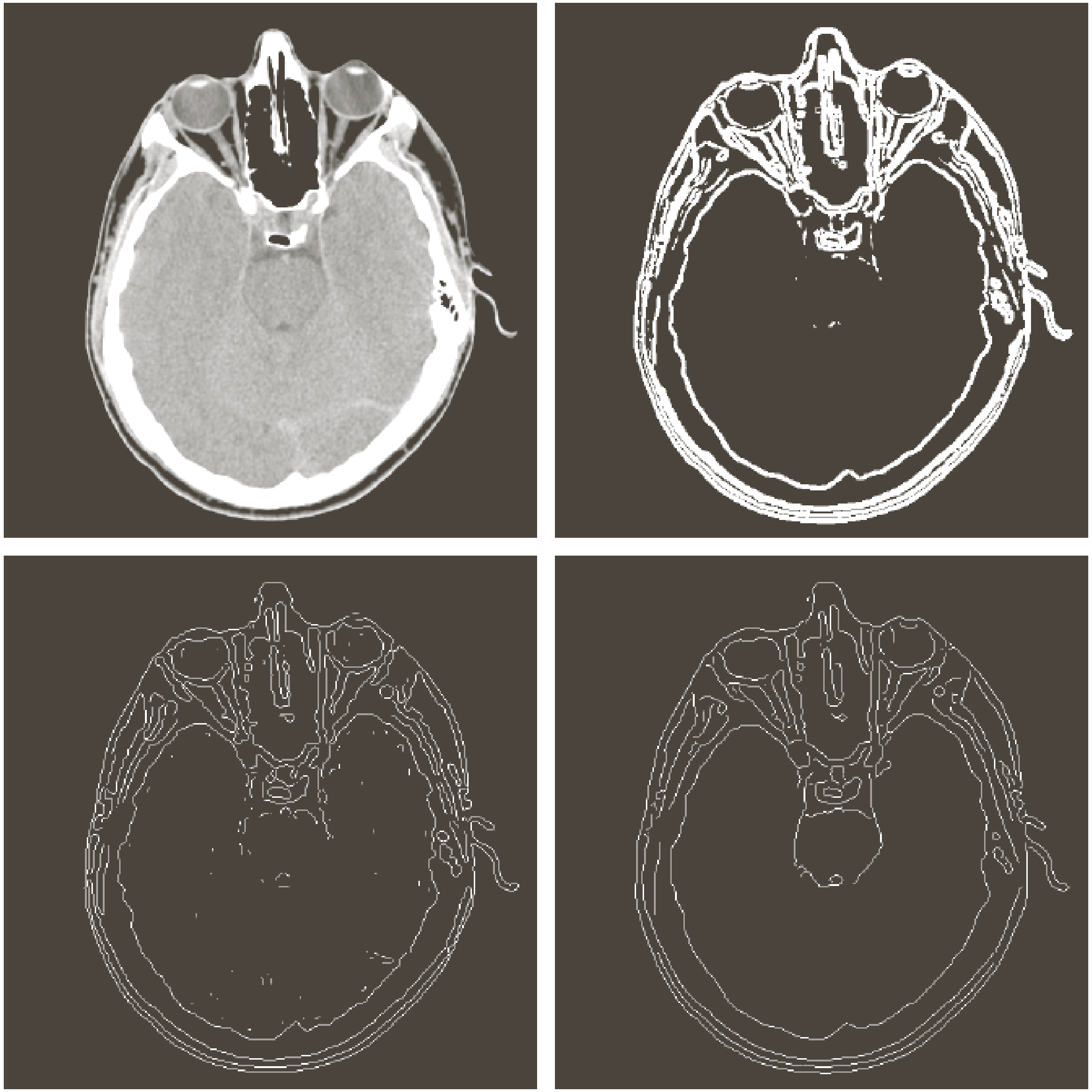
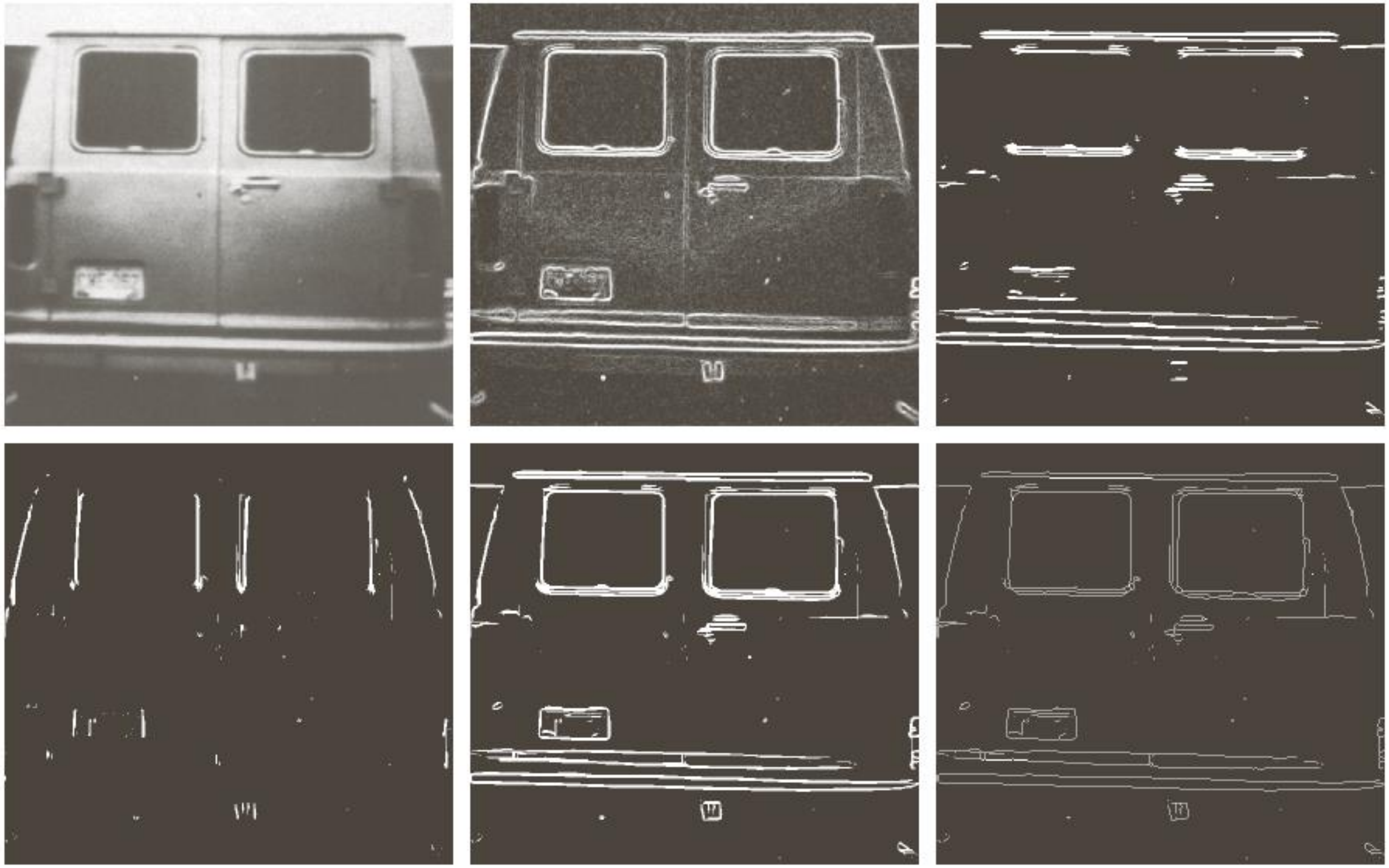


FIGURE 10.26 (a) Original head CT image of size 512 × 512 pixels, with intensity values scaled to the range [0, 1]. (b) Thresholded gradient of smoothed image. (c) Image obtained using the Marr-Hildreth algorithm. (d) Image obtained using the Canny algorithm. (Original image courtesy of Dr. David R. Pickens, Vanderbilt University.)

Boundary Detection—Local processing



a	b	c
d	e	f

FIGURE 10.27 (a) A 534 × 566 image of the rear of a vehicle. (b) Gradient magnitude image. (c) Horizontally connected edge pixels. (d) Vertically connected edge pixels. (e) The logical OR of the two preceding images. (f) Final result obtained using morphological thinning. (Original image courtesy of Perceptics Corporation.)

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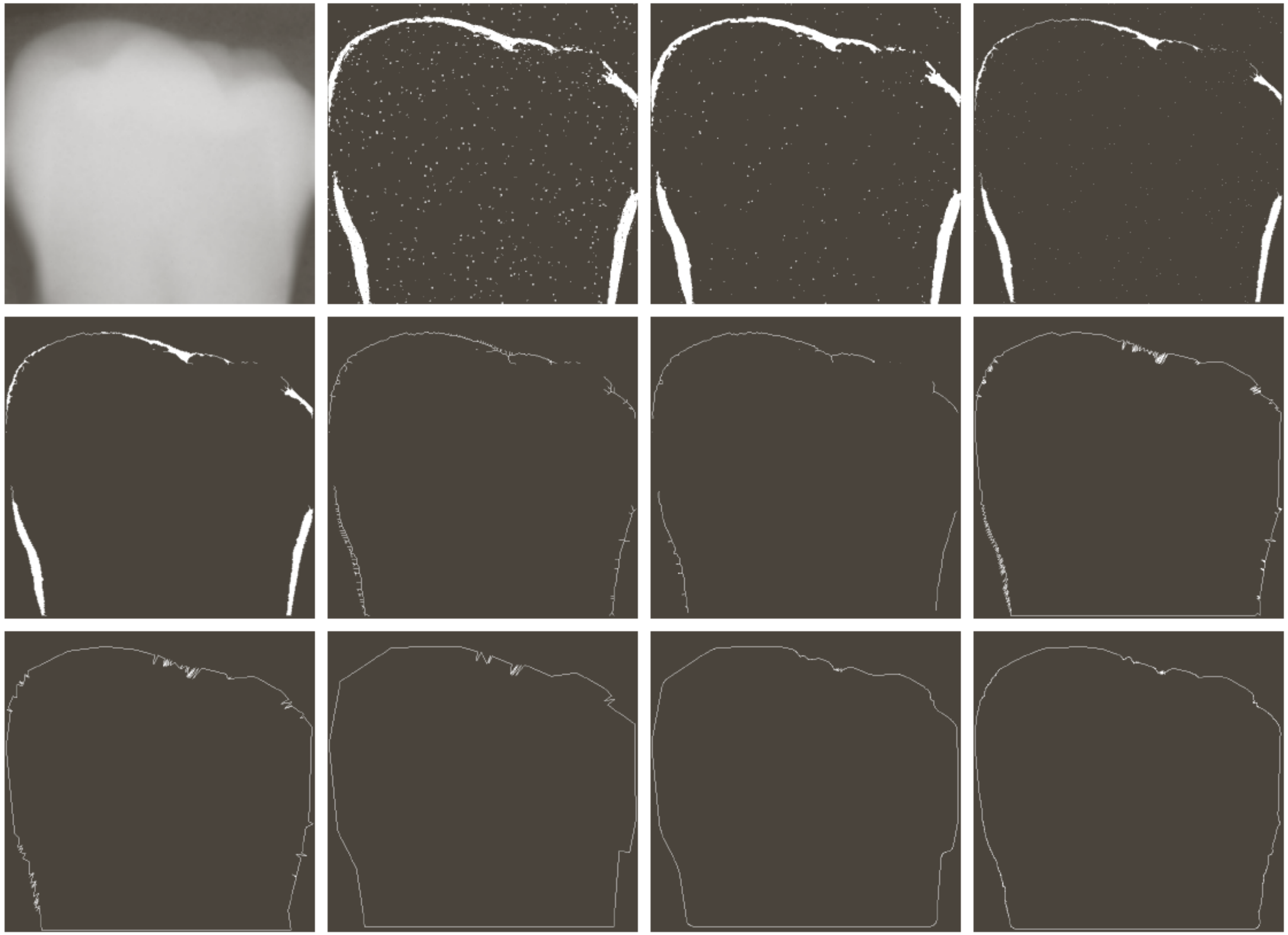
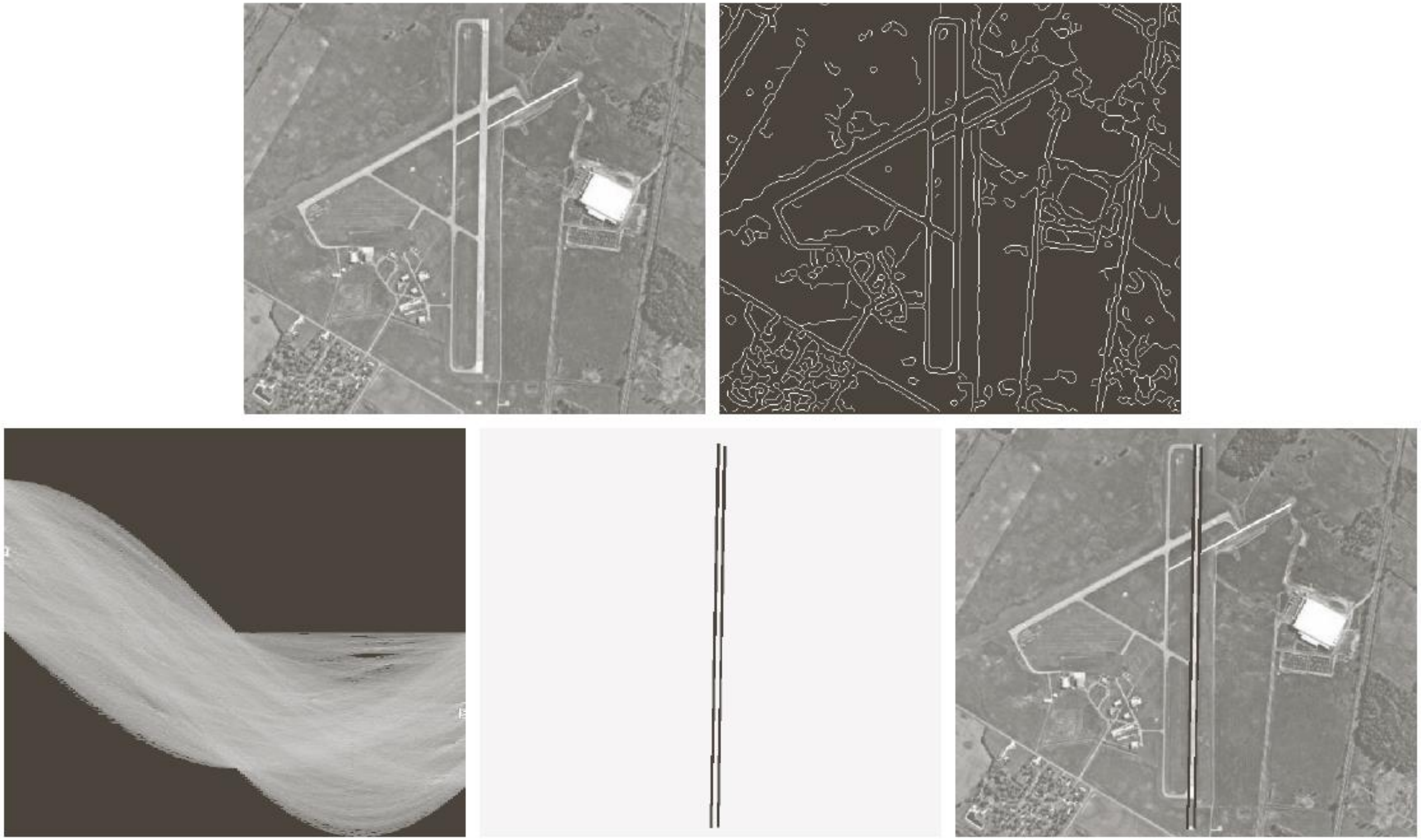
Boundary Detection—Regional processing



a	b	c	d
e	f	g	h
i	j	k	l

FIGURE 10.30 (a) A 550 × 566 X-ray image of a human tooth. (b) Gradient image. (c) Result of majority filtering. (d) Result of morphological shrinking. (e) Result of morphological cleaning. (f) Skeleton. (g) Spur reduction. (h)–(j) Polygonal fit using thresholds of approximately 0.5%, 1%, and 2% of image width ($T = 3$, 6, and 12). (k) Boundary in (j) smoothed with a 1-D averaging filter of size 1 × 31 (approximately 5% of image width). (l) Boundary in (h) smoothed with the same filter.

Boundary Detection—Hough transform



a b
c d e

FIGURE 10.34 (a) A 502 × 564 aerial image of an airport. (b) Edge image obtained using Canny's algorithm. (c) Hough parameter space (the boxes highlight the points associated with long vertical lines). (d) Lines in the image plane corresponding to the points highlighted by the boxes). (e) Lines superimposed on the original image.